



A Study of Starlink Decays during three Severe Geomagnetic Storms



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Abstract

The proliferation of mega-constellations in Low Earth Orbit (LEO) necessitates advanced orbital decay modelling to mitigate the risks of kinetic collisions and uncontrolled re-entries. Satellites in these regimes are subjected to thermospheric drag, a non-conservative force that modulates orbital longevity and is highly sensitive to solar-driven density fluctuations. This investigation utilizes Two-Line Element (TLE) datasets to quantify the impact of intense geomagnetic disturbances on the trajectory degradation of 48 Starlink satellites during the descending phase of Solar Cycle 25 (SC25). By extracting instantaneous state vectors, the study analyzes three major geomagnetic storms occurring between January 2025 and January 2026 to correlate storm duration with satellite decay. Results corroborate established aeronomic thresholds, confirming a critical transition altitude between 260 and 280 km where atmospheric density gradients precipitate non-linear descent rates. Furthermore, the analysis examines the cumulative perturbations on satellites that survived an initial storm in late 2025 only to succumb to subsequent disturbances in early 2026, suggesting a "pre-conditioning" effect on orbital stability. Observed trajectories indicate an inverse proportionality between geomagnetic storm duration and day differences (defined as the number of days spent between the reference altitude and the Kármán line). The heightened activity of SC25, exceeding the amplitude of its predecessor, provides a unique observational window into the drag dynamics of high-density satellite populations. These findings underscore the vulnerability of Very Low Earth Orbit (VLEO) architectures to solar-active conditions that were historically under-sampled. Ultimately, this work adds to the empirical evidence for the necessity of real-time space weather integration in mega-constellation traffic management systems

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Definitions

Before presenting the analysis method and the results, it is necessary to be familiar with some definitions of the important terms that will be encountered multiple times in this report. As the field of TLE analysis of satellite decays is relatively new, these definitions are not rigid.

The terms used in this report are largely based on the works of Oliveira et al. (2025).

1. Reference Altitude – 280 km. It can vary from 260 km to 320 km. In this work, it has been ensured that the reference altitude is chosen as close to the 280 km mark as possible, depending on data availability.
2. Reference Epoch – The epoch when the satellite crosses the reference altitude.
3. Re-entry Epoch – The epoch when the satellite crosses the Kármán line (100 km).
4. Predicted Re-entry Epoch – The epoch of re-entry predicted for the satellite when it is still at the reference altitude.
5. Prediction Error – The error, in days, between the predicted re-entry epoch and the actual re-entry epoch.
6. Day Difference – The number of days that the satellite takes to decay after reaching the reference altitude.
7. Average Decay Rate – Measured from the reference epoch to the re-entry epoch, in km/days.
8. Storm Intensity – Classified as mild ($\text{dst} < -50$ nT), moderate ($\text{dst} < -100$ nT) and severe ($\text{dst} < -200$ nT)
9. Quiet period – Uninterrupted period of several days when there are no geomagnetic storm conditions ($\text{dst} > -50$ nT)
10. Recovered Period – The period after storm peak (or equivalently, the minima in dst index) during which the dst index is increasing but has not reached > -50 nT

Methodology

The work begins by identifying severe storm conditions by looking at the dst index for each month, starting backwards from January 2026, as this work began in January 2026. For this, the data provided at wdc.kugi.kyoto-u.ac.jp by the World Data Centre is used.

Those months are chosen when the dst index fell below -200 nT. This work identified three most recent such events : January 2026, November 2025 and January 2025.

Next, the work utilizes the TLE (Two Line Element) data available at <http://www.space-track.org> to identify the satellites that decayed immediately after respective storms. By using the SATCAT tab on the website to check for Starlink decays, specifically those satellites were selected that are indicated to have decay within one week after the end of the recovery period.

Note that the definition of decay on the space-track website is “The lessening of the eccentricity of the elliptical orbit of an artificial satellite”. This happens before the satellite re-enters at 100 km.

Starlink satellites have been chosen for this work because they are the most abundant in number, and are found in LEO where the drag forces are highest. This also minimizes the differences in decays that can arise due to changes in geometry and other properties of the satellite.

Then, the ELSET Search tap is used to extract bulk TLE files for each satellite beginning at a common point in time. As the student targets a one-month window for TLE analysis, the TLE files are extracted for a one-month period centred at the storm minima.

Satellite analysis is performed by using the ephemeris package in python that can automatically calculate a satellite’s altitude based on the TLE parameters, once an observer is defined. The last available TLE file of the satellite is used to extrapolate the curve based on latest parameters till the satellite virtually crosses the Kármán line. The code for this work has been provided later in the report.

The algorithm is run for each satellite to obtain the desired parameters and plot individual trajectories.

Afterwards, the altitude decays of all the satellites corresponding to a particular storm are plotted with time against dst index values in the background for comparison. The dst values for each month are downloaded as text files from the earlier mentioned website, which are required to be manually parsed for use.

Results

The trajectory of STARLINK-5466 is shown as an example on the next page (Figure 1). Note the

commencement of the sharp decay after the satellite crosses reference altitude mark, indicated by the higher red dotted line.

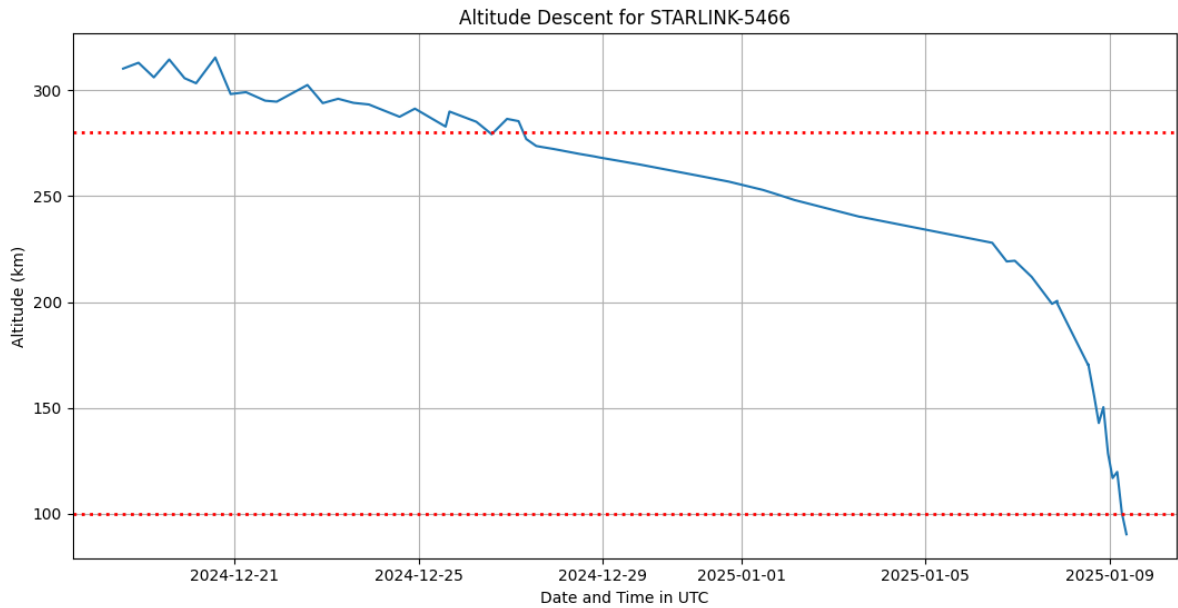


Figure 1 : Trajectory of SL-5466. The red dotted line between 250 km and 300 km indicates the reference altitude at 280 km. The second dotted line clearly marks the Kármán line.

Given below are the decay parameters for 10, 10 and 28 satellites decayed in January 2026, November 2025 and January 2025, respectively:

January 2026

Dst index minimum : -236 nT

Dst minimum epoch : 20 Jan 16:00 UT

Recovery Period ends : 22 Jan 19:00 UT

Quiet Period ends at : 19 Jan 20:00 UT

Reference Epoch : 278-282 km

Storm Duration = 51 hours

Satellite Name	Reference Epoch	Predicted Re-entry Epoch	Actual Re-entry Epoch	Prediction Error (in days)	Day Difference	Average Decay Rate (km/day)	Height during Quiet Period (km)
STARLINK-1705	2026/1/13 06:00:02	2026/1/30 06:00:02	2026/1/24 21:28:40	-5.355	11.645	15.457	247.919
STARLINK-2619	2026/1/13 04:00:05	2026/1/28 04:00:05	2026/1/24 09:53:26	-3.755	11.245	16.007	240.985
STARLINK-1700	2026/1/15 12:00:02	2026/2/4 12:00:02	2026/1/27 12:11:56	-7.992	12.008	14.99	262.626
STARLINK-30192	2026/1/21 00:00:03	2026/2/3 00:00:03	2026/1/29 22:02:26	-4.082	8.918	20.183	275.401
STARLINK-31227	2026/1/24 08:00:04	2026/1/30 08:00:04	2026/1/30 03:49:41	-0.174	5.826	30.895	322.916
STARLINK-1325	2026/1/21 00:00:03	2026/2/3 00:00:03	2026/2/1 08:01:54	-1.665	11.335	15.881	285.013

STARLINK-1740	2026/1/20 00:31:16	2026/1/30 00:31:16	2026/1/30 03:57:21	0.143	10.143	17.746	274.359
STARLINK-2117	2026/1/20 00:00:03	2026/2/4 00:00:03	2026/2/1 15:05:17	-2.371	12.629	14.253	277.199
STARLINK-30228	2026/1/24 04:00:01	2026/2/7 04:00:01	2026/2/3 13:32:46	-3.602	10.398	17.311	286.342
STARLINK-35238	2026/1/25 14:00:01	2026/1/29 14:00:01	2026/2/2 00:36:23	3.442	7.442	24.187	351.189

Average day difference = 10.159 days

Average decay rate = 18.691 km/day

November 2025

Dst index minimum : -217 nT

Dst minimum epoch : 12 Nov 04:00 UT

Recovery Period ends : 13 Nov 14:00 UT

Quiet Period ends at : 5 Nov 23:00 UT

Reference Epoch : 277-283 km

Storm Duration = 34 hours

<u>Satellite Name</u>	<u>Reference Epoch</u>	<u>Predicted Re-entry Epoch</u>	<u>Actual Re-entry Epoch</u>	<u>Prediction Error (in days)</u>	<u>Day Difference</u>	<u>Average Decay Rate (km/day)</u>	<u>Height during Quiet Period (km)</u>
STARLINK-33667	2025/11/8 00:55:46	2025/11/22 00:55:46	2025/11/17 11:45:36	-4.549	9.451	19.045	233.791
STARLINK-3846	2025/11/2 08:00:02	2025/11/11 08:00:02	2025/11/16 13:33:30	5.232	14.232	12.648	267.698
STARLINK-1010	2025/10/30 08:00:02	2025/11/25 08:00:02	2025/11/16 16:16:47	-8.655	17.345	10.378	262.815
STARLINK-1903	2025/11/7 04:00:03	2025/11/22 04:00:03	2025/11/17 01:39:07	-5.098	9.902	18.178	297.011
STARLINK-3183	2025/11/11 00:00:03	2025/11/20 00:00:03	2025/11/17 20:51:31	-2.131	6.869	26.204	336.603
STARLINK-5008	2025/11/4 04:00:02	2025/11/22 04:00:02	2025/11/18 10:11:56	-3.742	14.258	12.624	269.561
STARLINK-1292	2025/11/6 08:00:02	2025/11/24 08:00:02	2025/11/20 02:13:39	-4.241	13.759	13.082	280.064
STARLINK-2651	2025/11/8 04:19:31	2025/11/27 04:19:31	2025/11/20 02:10:09	-7.09	11.91	15.113	294.211
STARLINK-32357	2025/11/15 00:49:40	2025/11/30 00:49:40	2025/11/28 17:10:48	-1.319	13.681	13.157	
STARLINK-1054	2025/11/8 08:00:03	2025/11/25 08:00:03	2025/11/21 23:04:00	-3.372	13.628	13.208	290.77

Average day difference = 12.504 days

Average decay rate = 15.364 km/day

January 2025

Dst index minimum : -212 nT

Dst minimum epoch : 1 Jan 15:00 UT

Recovery Period ends : 2 Jan 23:00 UT

Quiet Period ends at : 1 Jan 08:00 UT

Reference Epoch : 276-284 km

Storm Duration = 32 hours

<u>Satellite Name</u>	<u>Reference Epoch</u>	<u>Predicted Re-entry Epoch</u>	<u>Actual Re-entry Epoch</u>	<u>Prediction Error (in days)</u>	<u>Day Difference</u>	<u>Average Decay Rate (km/day)</u>	<u>Height during Quiet Period (km)</u>
STARLINK-2020	2024/12/25 04:00:01	2025/1/9 04:00:01	2025/1/4 18:55:12	-4.378	10.622	16.947	242.318
STARLINK-5784	2024/12/21 22:00:02	2025/1/1 22:00:02	2025/1/4 20:49:14	2.951	13.951	12.902	195.204
STARLINK-1486	2024/12/21 06:00:01	2025/1/9 06:00:01	2025/1/5 03:29:58	-4.104	14.896	12.084	233.334
STARLINK-2152	2024/12/26 06:00:01	2025/1/11 06:00:01	2025/1/6 01:02:19	-5.207	10.793	16.677	255.255
STARLINK-3666	2024/12/21 22:00:02	2025/1/1 22:00:02	2025/1/5 12:06:00	3.587	14.587	12.339	253.065
STARLINK-3684	2024/12/23 22:00:01	2025/1/2 22:00:01	2025/1/5 08:14:00	2.426	12.426	14.485	237.138
STARLINK-5404	2024/12/22 06:00:01	2025/1/4 06:00:01	2025/1/5 00:01:47	0.751	13.751	13.09	222.7
STARLINK-1528	2024/12/28 12:00:02	2025/1/8 12:00:02	2025/1/5 18:04:30	-2.747	8.253	21.81	247.264
STARLINK-2244	2024/12/30 00:00:02	2025/1/10 00:00:02	2025/1/5 07:14:00	-4.699	6.301	28.565	257.313
STARLINK-4009	2024/12/24 22:00:01	2025/1/4 22:00:01	2025/1/5 14:36:16	0.692	11.692	15.395	238.132
STARLINK-1720	2024/12/27 22:00:01	2025/1/11 22:00:01	2025/1/7 00:08:20	-4.911	10.089	17.841	270.261
STARLINK-2347	2024/12/29 14:00:01	2025/1/9 14:00:01	2025/1/6 17:37:54	-2.849	8.151	22.082	266.132
STARLINK-1602	2024/12/30 14:00:02	2025/1/11 14:00:02	2025/1/8 20:12:13	-2.742	9.258	19.442	259.47
STARLINK-2181	2024/12/30 10:00:01	2025/1/10 10:00:01	2025/1/8 12:01:53	-1.915	9.085	19.814	267.132
STARLINK-2392	2024/12/29 22:00:01	2025/1/9 22:00:01	2025/1/7 09:14:03	-2.532	8.468	21.256	264.719
STARLINK-2441	2024/12/30 10:00:01	2025/1/10 10:00:01	2025/1/7 23:24:27	-2.441	8.559	21.031	266.763
STARLINK-3886	2024/12/26 06:00:01	2025/1/6 06:00:01	2025/1/8 01:06:47	1.796	12.796	14.066	247.572
STARLINK-2546	2024/12/31 04:00:01	2025/1/12 04:00:01	2025/1/9 20:37:54	-2.307	9.693	18.57	270.284
STARLINK-2541	2024/12/31 16:00:01	2025/1/10 16:00:01	2025/1/10 07:03:50	-0.372	9.628	18.696	267.412
STARLINK-2477	2024/12/31 08:00:01	2025/1/11 08:00:01	2025/1/9 23:27:11	-1.356	9.644	18.665	279.115
STARLINK-2573	2024/12/31 20:00:01	2025/1/11 20:00:01	2025/1/10 05:14:51	-1.615	9.385	19.179	272.733

STARLINK-2688	2024/12/30 14:00:02	2025/1/10 14:00:02	2025/1/10 02:34:29	-0.476	10.524	17.104	267.041
STARLINK-4003	2024/12/27 04:00:02	2025/1/7 04:00:02	2025/1/8 18:29:18	1.604	12.604	14.282	246.914
STARLINK-4601	2024/12/26 14:00:01	2025/1/14 14:00:01	2025/1/9 17:57:21	-4.835	14.165	12.708	254.151
STARLINK-1891	2025/1/1 06:00:01	2025/1/13 06:00:01	2025/1/9 15:55:38	-3.586	8.414	21.394	280.269
STARLINK-5144	2024/12/26 06:00:01	2025/1/15 06:00:01	2025/1/9 22:00:02	-5.333	14.667	12.273	266.674
STARLINK-5466	2024/12/26 14:00:01	2025/1/14 14:00:01	2025/1/9 08:42:00	-5.221	13.779	13.063	252.897
STARLINK-30406	2025/1/6 10:00:01	2025/1/9 10:00:01	2025/1/9 02:59:27	-0.292	2.708	66.471	352.734

Average day difference = 10.675 days

Average Decay Rate = 19.008 km/day

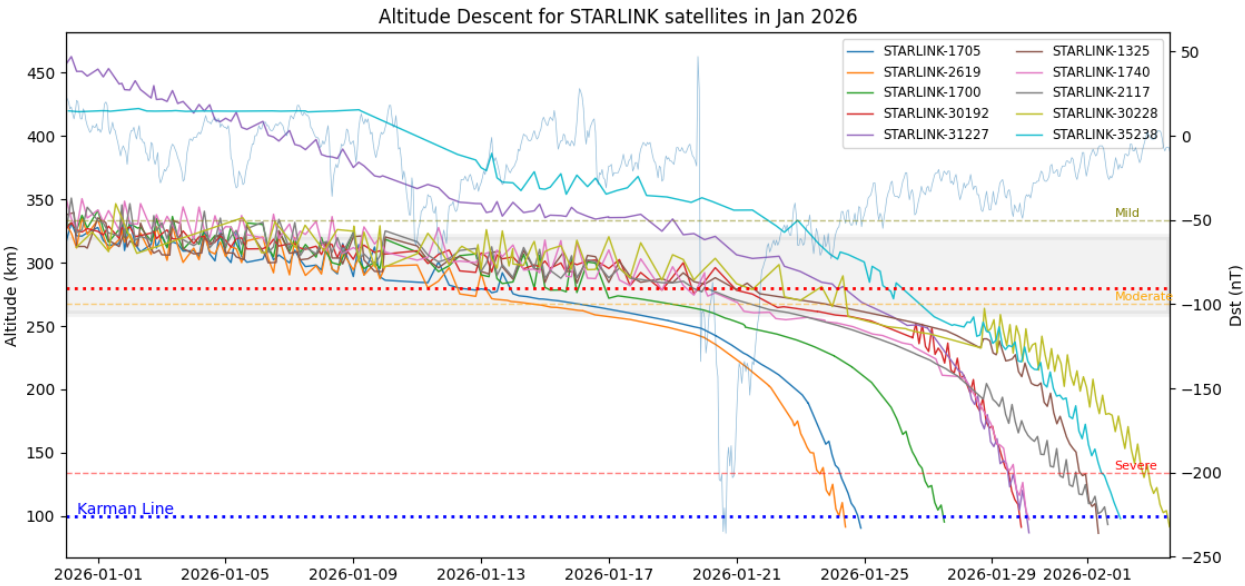


Figure 2 : The decays of Starlink satellites during the January 2026 storm

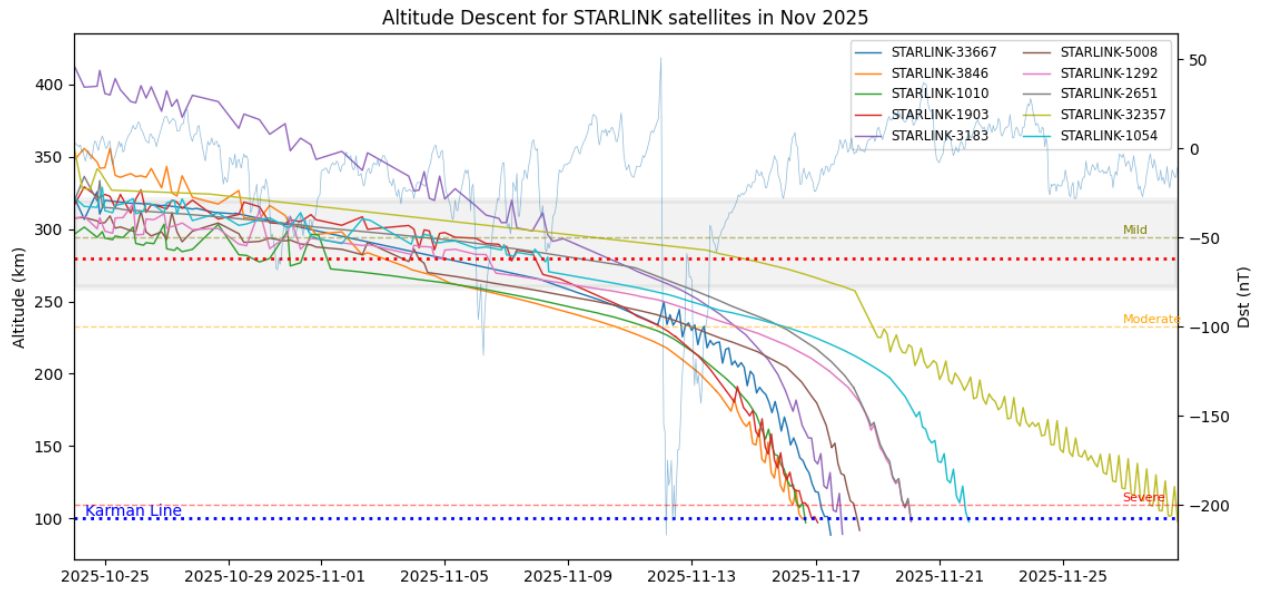


Figure 3 : The decays of Starlink satellites during the November 2025 storm

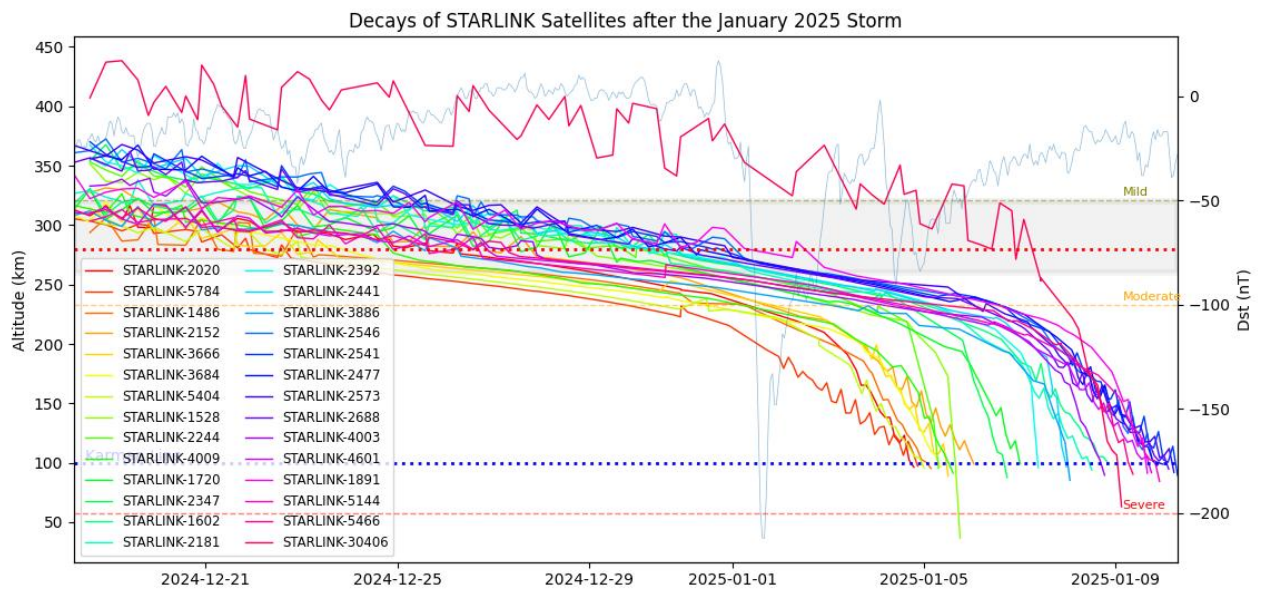


Figure 4 : The decays of Starlink satellites during the January 2025 storm

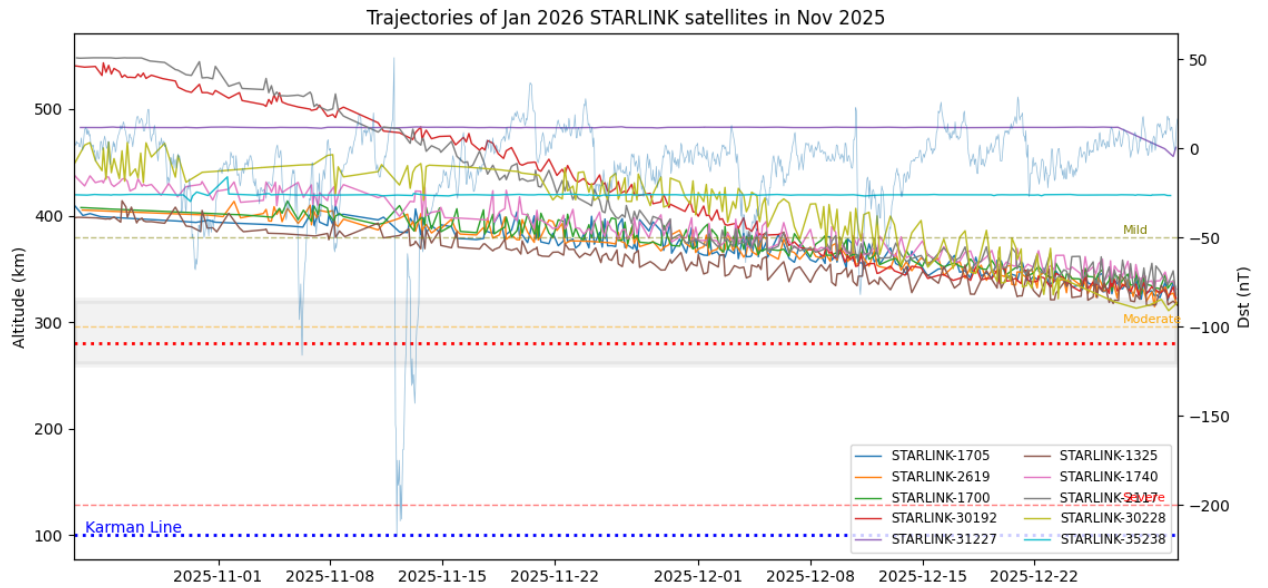


Figure 5 : Effect of the November 2025 storm on the satellites that decayed later in January 2026. Note that the satellites have descended by about a hundred kilometers, but their decay is not apparent.

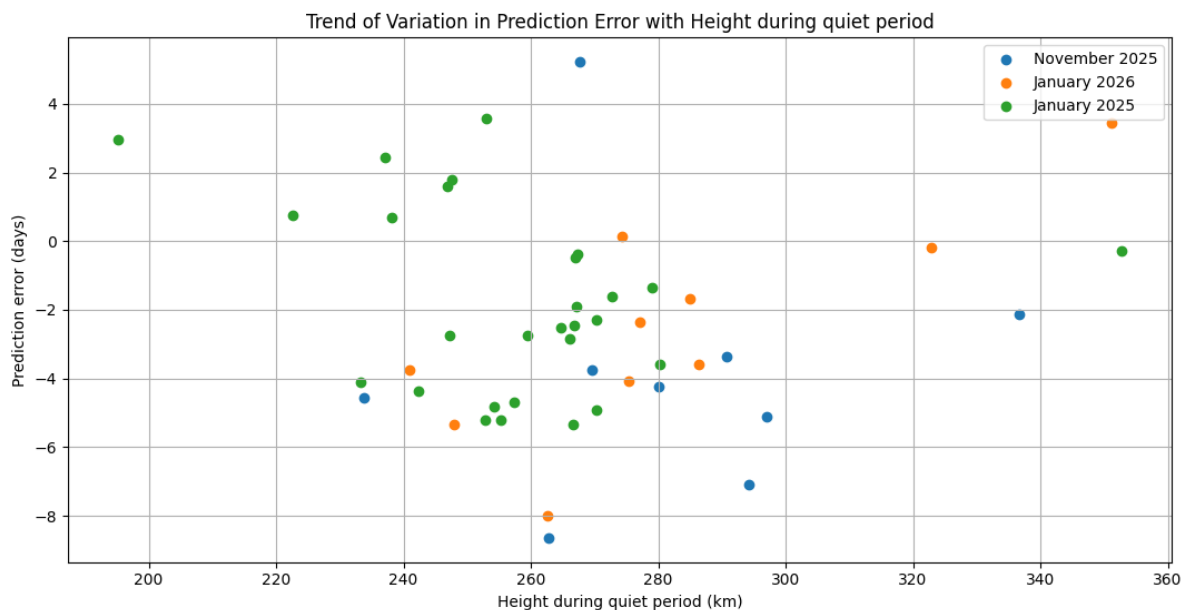


Figure 6 : Correlation plot between the prediction error and height of a satellite during the quiet period. No correlation has been found (coefficient of correlation = 0.0007)

Conclusion

It has been clearly shown that the occurrence of a geomagnetic storm accelerates the orbital descent of satellites. For longer recovery periods (January 2026), the day difference also reduces, indicating that the prolonged storm forced the satellite to re-

enter before it would have, had the storm occurred for a shorter duration.

It was expected that a higher altitude during the quiet period would have led to the decay of satellite after the predicted date and thus, resulting in a positive prediction error. But this is largely influenced by the choice of quiet period.

For the November 2025 and January 2026 storms combined, the correlation coefficient has been

calculated as 0.417. But if the data from the January 2025 storm is added too, the coefficient becomes ~0.

This is because there was negligible solar activity in December 2024, ending the quiet period just before the storm.

But a consistent observation is that satellites orbiting at higher altitudes just before the storm re-enter later and have less steeper decays.

Future Scope

To establish clear correlation between storm duration and day difference, analysis on larger data set with varying storm intensities is required.

The work done by the student was largely manual in terms of acquiring data and deriving decay parameters. Subsequent work on larger dataset needs automation of the process.

Code

<https://github.com/SwastikaSanghi/Analyzing-Effect-of-Geomagnetic-Activity-on-Starlink-Decays>

AI Usage Statement

Generative AI has been used only to generate the image for the cover of the report.

References

- 1) "Tracking reentries of Starlink satellites during the rising phase of solar cycle 25" Oliveira et al. (2025)
- 2) "The 10 October 2024 geomagnetic storm may have caused the premature reentry of a Starlink satellite" Oliveira et al. (2025)
- 3) "Loss of 12 Starlink Satellites Due to Intense Space Weather Activity Surrounding the Extreme Geomagnetic Storm of 10 May 2024" Ashruf et al. (2025)